Last time

n-simplex Δ(S) — convex hull in R^N of a set S of size n + 1 in general pos

subsimplex — Δ(S′) for some S′ sub S

boundary ∂Δ(S) — union of proper subsimplices

interior Int(Δ(S)) — Δ(S) – ∂Δ(S)

simplicial complex — K = {Δ(S)}_{S in I} s.t.

1) the Int(Δ(S)) partition bigcup_{S in I} Δ(S)
2) if S in I and S′ sub S, then S′ in I too

|K| = bigcup_{S in I} Δ(S) is the realization of K

today: instead of Δ({v0, v1, …, vk})
we will write [v0, v1, …, vk] [for convenience]

Ex — pick a, b, c in R^3

[which are simplicial complexes?]

{[a], [b], [c], [a, b], [b, c]}
{[a], [b], [c], [a, b]}
{[a], [c], [a, b]}

Ex — pick p < q < r in R

[which are simplicial complexes?]

{[p], [q], [r], [p, q], [p, r]}
{[p], [q], [r], [p, q], [q, r]}
{[p], [q], [r], [p, r]}

Euler characteristic of K:

$$\chi(K) = e_0(K) - e_1(K) + e_2(K) - \dots$$

where $e_n(K)$ = # of n-simplices in K

Ex — {[p], [q], [r], [p, q], [q, r]} has Euler char 3 – 2 = 1

cf. Problems 67–68

Ex — simplicial tetrahedron has Euler char 4 – 6 + 4 = 2 [cf. Euler's formula]

Thm — if $|K_1| \sim |K_2|$, then $\chi(K_1) = \chi(K_2)$

Q — is the converse true?

A — no! tetrahedron vs {0, 1}

(Crossley §9.1) — goal: find an improvement of χ that can distinguish tetrahedron vs {0, 1}

idea #1 — instead of a single number χ, use the sequence $e_0, e_1, e_2, \dots$

e_i's not invariant under homotopy equivalence!

idea #2 — modify the e_i's to get an invariant sequence $b_0, b_1, b_2, \dots$

Ex $K_1 = \{[a], [b], [c], [a, b]\}$
$K_2 = \{[a], [c]\}$

then $|K_1| \sim |K_2|$
$(e_i(K_1))_i = (3, 1, 0, 0, \ldots)$
$(e_i(K_2))_i = (2, 0, 0, 0, \ldots)$

notice: $\partial\Delta(\{a, b\}) = \Delta(\{a\}) \cup \Delta(\{b\})$

idea #3 use ∂ to encode how the segment between a and b contracts down to a

Chains with Z/2Z Coefficients

Df let $S_n(K)$ be the set of n-simplices in K
let $C_n(K)$ be the power set of $S_n(K)$

we can view $C_n(K)$ as an abelian group!

write any elt of $C_n(K)$ as a formal sum: e.g.,
if $c = \{\Delta_1, \Delta_2, \Delta_3\}$ in $C_n(K)$
then as a formal sum, $c = \Delta_1 + \Delta_2 + \Delta_3$

the group law is addition mod 2: e.g.,
$(\Delta_1 + \Delta_2 + \Delta_3) + (\Delta_3 + \Delta_4)$
$= \Delta_1 + \Delta_2 + \Delta_4$

Rem as a formal sum, $\varnothing$ becomes 0

Rem since $c + c = 0$ for all c in $C_n(K)$
can view $C_n(K)$ as a $\mathbb{Z}/2\mathbb{Z}$-vector space

elts of $C_n(K)$ are called Z/2Z simplicial chains any n-simplex Δ in K can now be viewed as a simplicial chain Δ in C_n(K)

<u>Df</u> the <u>boundary chain</u> of Δ is

$$\partial\Delta = \mathrm{sum}_{\text{faces } \Delta'} \Delta' \text{ in } C_{n-1}(K)$$

[recall: faces are maximal proper subsimplices]

<u>Ex</u> ∂[a, b, c] = [a, b] + [a, c] + [b, c]

the <u>boundary map</u> ∂_n : C_n(K) to C_{n – 1}(K) is defined as follows [“by linearity”]:

if c = sum_i Δ_i
then ∂c = sum_i ∂Δ_i

<u>Df</u> let Z_0(K) = C_0(K)
let Z_n(K) = ker(∂_n) for n > 0
let B_n(K) = im(∂_{n + 1})

elts of Z_n(K) are called <u>Z/2Z cycles</u>
elts of B_n(K) are called <u>Z/2Z boundaries</u>

<u>Ex</u> K = {[a], [b], [c], [a, b]}
C_0 gen over Z/2Z by [a], [b], [c]
C_1 gen over Z/2Z by [a, b]

∂ : C_1 to C_0 is def by ∂[a, b] = [a] + [b]
Z_0 gen over Z/2Z by [a], [b], [c]
Z_1 is trivial
B_0 gen over Z/2Z by [a] + [b]
B_1 is trivial

Df the nth mod 2 simplicial homology group of K is

$$H_n(K) = Z_n(K)/B_n(K)$$

the nth mod 2 Betti number of K is

$$b_n = \dim_{\mathbb{Z}/2\mathbb{Z}} H_n(K)$$

i.e., size of any minimal generating set for $H_n(K)$

Ex $K = \{[a], [b], [c], [a, b]\}$

$b_0 = 2$

$b_1 = 0$